



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Assignment – 2
CSE2218 Section A Summer 2025

Note: Please follow the sample input and output format. Otherwise, your marks will be deducted.

1	You are given an undirected connected weighted graph with n vertices and m edges. Each edge has a weight representing its cost. Find the Second Best Minimum Spanning Tree (MST). Input: The first line contains two integers: n (number of vertices) and m (number of edges). The next m line each contains three integers: u v w (two endpoints of the edge and its weight). Here m > n and the graph is connected, also there are multiple spanning trees in the given inputs. Output: Print the second best MST weight. Then in the next lines, print the edges for the second best MST. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>3 3 0 1 1 1 2 1 0 2 2</td><td>Second Best MST Weight: 3 Edges in Second Best MST: 0 - 1 (1) 0 - 2 (2)</td></tr><tr><td>4 5 0 1 1 1 2 2 0 2 3 2 3 1 1 3 4</td><td>Second Best MST Weight: 5 Edges in Second Best MST: 0 - 1 (1) 0 - 2 (3) 2 - 3 (1)</td></tr></table> Hint: Use Kruskal's Algorithm first to get the minimum spanning tree. Then remove one edge from the minimum spanning tree, and add one edge previously not added to build a new spanning tree. From the newly created spanning trees, choose the best one.	Sample Input	Sample Output	3 3 0 1 1 1 2 1 0 2 2	Second Best MST Weight: 3 Edges in Second Best MST: 0 - 1 (1) 0 - 2 (2)	4 5 0 1 1 1 2 2 0 2 3 2 3 1 1 3 4	Second Best MST Weight: 5 Edges in Second Best MST: 0 - 1 (1) 0 - 2 (3) 2 - 3 (1)	10
Sample Input	Sample Output							
3 3 0 1 1 1 2 1 0 2 2	Second Best MST Weight: 3 Edges in Second Best MST: 0 - 1 (1) 0 - 2 (2)							
4 5 0 1 1 1 2 2 0 2 3 2 3 1 1 3 4	Second Best MST Weight: 5 Edges in Second Best MST: 0 - 1 (1) 0 - 2 (3) 2 - 3 (1)							
2	You are given a connected weighted directed graph with n vertices (numbered 0 to n-1) and m edges. Each edge connects two vertices u and v with a weight w. You are also given three special vertices: s (source node), t (destination node), and k (mandatory node). You must find the shortest path from s to t that passes through the node k at least once. If there is no valid path, print "No valid path". Input: The first line contains two integers: n (number of vertices) and m (number of edges). The next m line each contains three integers: u v w (two endpoints of the edge and its weight). The last line contains three integers: s (source node), t (destination node), and k (mandatory node).	10						

Output: If there is no valid path, print “No valid path”. Otherwise, first **print the shortest distance and then print the path in the proper format given.**

Sample Input	Sample Output
6 7 0 1 4 0 2 2 1 2 5 1 3 10 2 4 3 4 3 4 3 5 11 0 5 2	Shortest distance from 0 to 5 passing through 2: 20 Path: 0 -> 2 -> 4 -> 3 -> 5
5 4 0 1 2 1 2 3 3 4 5 0 4 10 0 4 2	No valid path

Hint: Please note that the **graph is directed, not undirected**, so make your adjacency list accordingly. First, write a **function named dijkstra** that takes **three parameters: number of vertices, adjacency list and source node** and **returns the distance vector and parent vector (for path reconstruction)**. You need to **calculate the minimum distance from source node to mandatory node using dijkstra**, then you need to **calculate the minimum distance from mandatory node to destination node using dijkstra**. Use the **same dijkstra function twice, just changing the value of source node**. Then **add the distance and reconstruct the path** from the parent vector.